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ORIGINAL CONTRIBUTION

Shear and extensional rheology of polyethylenes recycled using a solvent dissolution process

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Abstract
There is high market demand for increasing the viability of current plastic recycling processes. In this work rheology is used to evaluate the mechanical properties of a solvent dissolution recycled polymer compared to its virgin untreated precursor. Solvent dissolution and precipitation are used to target multi-layer, multi-component, industry films, which cannot be mechanically recycled. Polyethylene was chosen as the primary polymer of interest. Polymer thermal stability was monitored via time-resolved rheology; consecutive frequency sweeps over the course of an hour while under isothermal conditions. Additional rheological experiments were performed within the identified thermally stable conditions. Small-angle oscillatory shear was complemented with steady shear viscosity experiments over a wide range of shear rates. Extensional rheology was used to determine changes in molecular weight and cross link density. Rheological characterization is supplemented with gas chromatography–mass spectrometry of the solvent wash to determine components stripped from the virgin polymers during solvent treatment.

Keywords Plastic recycling · Rheology · Recycled polymer · Solvent dissolution

Introduction

Less than 14% of plastics are currently recycled; roughly 40% is destined for landfills, 14% incinerated, and 32% ends up leaking into the environment (Geyer et al. 2017; Li et al. 2022). Environmental concerns associated with the large-scale usage of plastics have motivated significant research into advancing plastic recycling. The oldest form of plastic recycling is mechanical recycling (Jehanno et al. 2022). The process starts with a grinding step, followed by compounding and shaping into pellets. However, this type of processing has been shown to cause degradation of the polymer's structure (Pillin et al. 2008; Curtzwiler et al. 2011; Bahlouli et al. 2012). Several other newer approaches are currently being commercialized including chemical recycling, such as catalytic hydrogenolysis (Sánchez-Rivera and Huber 2021), PET methanolysis (Li et al. 2022), or pyrolysis (Qureshi et al. 2020; Zhao et al. 2020). Chemical recycling aims to break down the polymer molecules into monomer constituents and/or fine chemicals. Another recycling approach uses solvents to remove contaminants or individual polymers from a multicomponent plastic (Zhao et al. 2018; Ügdüler et al. 2020). These newer advancements in recycling often handle certain challenges better than mechanical recycling, such as contamination with food or adhesives, the diversity of additives and processing aids (Wiesinger et al. 2021), and multi-component systems like multilayer films used in packaging.

Plastics' largest market sector is packaging, accounting for approximately 42% of all non-fiber plastics usage (Geyer et al. 2017). Many of these packaging materials are multi-layer films composed of multiple polymer types. Each layer provides new functionality while simultaneously reducing recyclability. Mechanical recycling is incapable of processing multilayer films, because the different polymers are not chemically compatible leading to phase separation during processing (Kaiser et al. 2018). The representative multi-layer packaging film for this study is a commercial three-layer system from Amcor using polyethylene (PE), ethylene vinyl alcohol (EVOH), and polyethylene terephthalate

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(PET). PE acts as a flexible moisture barrier and is used in different tie layers, EVOH as an oxygen barrier, and PET provides rigidity and durability. This film is discussed in more detail by Sánchez-Rivera et al. (2021). Within their study the representative film is processed using solvent targeted recovery and precipitation (STRAP). The film is dissolved in multiple solvents at elevated temperatures to separate the film and isolate the individual pure polymers. Once dissolved, each of the polymers is selectively precipitated out of solution by either adding an antisolvent (Walker et al. 2020) or by reducing the temperature (Sánchez-Rivera et al. 2021). The precipitated polymers are then filtered out of the solution one at a time to recover the pure polymer from the film. Solvent dissolution has thus far proven to be the most promising method of recycling multilayer films such as the Amcor film described above (Cervantes-Reyes et al. 2015; Ügdüler et al. 2020; Walker et al. 2020; Sánchez-Rivera et al. 2021; Zhou et al. 2021) and has very recently seen commercialization within industry (Ügdüler et al. 2020). Three major industry examples of solvent based recycling methods are APK's Newcycling chemical dissolution process (APK 2024), Purecycle's purification process for upcycling polypropylene (Purecycle 2024), and the CreaSolv process developed by CreaCycle GmbH and the Fraunhofer Institute IVV (Fraunhofer 2024). Most industry solvent-based recycling processes can be considered early commercial or pilot scale, so while there is significant interest in this methodology, further work is required to enable wider distribution of the technique.

STRAP was initially applied on rigid multilayer films containing primarily polyethylene, ethylene vinyl alcohol (EVOH), and polyethylene terephthalate (PET) (Walker et al. 2020). The purpose of this study was to investigate how solvent dissolution methodologies such as STRAP affect polymer film properties and microstructures using a combination of differential scanning calorimetry (DSC), shear rheology, and extensional rheology measurements. Shear and extensional rheology are highly sensitive to changes in polymer morphology, molecular weight distribution, crosslinking, and branching. As a result, shear and extensional rheology measurements can be excellent indicators of a polymer's processibility and its potential end uses after recycling. To simplify this study, rather than beginning with a multilayer film, virgin PE was put through the STRAP process and measurements were taken before and after solvent treatment. This approach avoids potential contamination from other polymers found in an actual multicomponent film. By eliminating the possibility of residual secondary polymers, the effect of solvent treatment alone on the rheological properties of the virgin polymers can be more accurately determined. Low- and high-density polyethylenes (LDPE and HDPE) were chosen due to their prevalence in the single-use packaging market (Jehanno et al. 2022) and their use in the representative Amcor multi-layer film (Sánchez-Rivera et al. 2021). Subsequent work will determine the impact of solvent processing on the rheological properties of the other two layers, and after that ultimately the impact on the actual recycled film.

In this paper, the effect of solvent dissolution on the rheological properties of two different polyethylene (PE) polymers is studied. In the second section, the techniques and methods are described including the sample preparation, the shear and extensional rheometer set-ups, and a time-resolved mechanical spectroscopy measurement designed to quantify the stability of each polymer. In the third section, the results are presented. We show that solvent treatment has a negative impact on overall stability of the polymers and quantitatively alters the shear and extensional rheology. In the fourth section, the experimental results are discussed in more detail and conclusions are drawn.

Experimental methods

Materials

For this study, two polyethylene (PE) polymers were used: 608A low-density polyethylene (LDPE) from DOW having a melt flow index of 2.6 g/10 min and a density of 0.923 g/cm^3, and HP-167-AB high-density polyethylene (HDPE) from NOVA Chemicals having a melt flow index of 1.2 g/10 min and a density of 0.967 g/cm^3. The rheology of each of these virgin samples was studied without further processing to establish a baseline for comparison with recycled polymers. Gel permeation chromatography (GPC) of the PEs is included in the supplemental information, Figs. S1 and S2.

STRAP experimental procedure

The LDPE and HDPE pellets were subjected to the solvent-targeted recovery and precipitation (STRAP) process as outlined below to elucidate the effects that solvent-based recycling processes can have on the melt rheology and the processability of the recycled polymers. The solvent used for PE in the STRAP procedure was toluene. In a single experiment, the LDPE and HDPE pellets were added to toluene, in a 1 to 7.5 ratio, at 110 °C, and allowed to dissolve under constant stirring for 1 h. After the dissolution was completed, the solution was removed from the heating source and the PE was allowed to precipitate using an ice water bath for 10 min. The solvent was separated from the precipitated polymer via vacuum filtration for 15 min at − 0.7 bar. The precipitated polymer was then dried in a vacuum oven for 3 h at − 1 bar and 100 °C. The final product was lightweight flakes or chunks that could be melt pressed directly without conversion into pellets.

Gas chromatography–mass spectrometry

Gas chromatography–mass spectrometry (GC-MS) was used to identify unknown compounds in the solvents after polymer dissolution following the STRAP process. A Shimadzu GCMS-QP2010S with a DB-5 column (Agilent) was used at a maximum oven temperature of 310 °C. For each analysis, a 1 μL solvent sample was injected and the initial oven temperature was held at 40 °C for 1 min. The oven temperature was increased to 100 °C at a ramping rate of 5 °C/min, then increased to 310 °C at a ramping rate 10 °C/min, and held at this temperature for 15 min. During this analysis, the vaporized components were transported through the DB-5 column and these underwent separation based on their distinct chemical properties, leading to individual elution. As these separated components exited the column, they entered the mass spectrometer, which ionizes the samples to provide the mass to charge (m/z) ratios and compare them to known compounds in available libraries. In this system, helium was the carrier gas at a total flow rate of 8.7 mL/min.

A high-temperature gas chromatography system with a flame ionization detector (GC-FID), model Shimadzu GC-2010, was used to quantify the linear alkane concentrations, from C8 to C40. The corresponding linear alkane standards were prepared in the concentration range of 0 to 500 ppm. This GC-FID was equipped with a Restek MXT-1HT column. For each measurement, the initial oven was 40 °C and this was held for 5 min. The oven temperature was increased to 415 °C at a ramping rate of 6 °C/min and held at this temperature for 6.15 min. Hydrogen was the carrier gas in this system, at a total flow rate of 40 mL/min.

Melt press sample preparation

In preparation for shear and extensional testing, HDPE and LDPE samples were formed into disks (and cylinders for the filament stretching) using a high-temperature vacuum press. The vacuum is essential to prevent oxidation of the polymers at elevated temperatures. The procedure for melt pressing the sample disks (or cylinders) was as follows: First, the polymer pellets or flakes were placed into the stainless steel mold of the desired size and shape, and sandwiched between Teflon and aluminum sheets. The use of Teflon allowed for easy release of the polymer samples from the mold after formation. The polymer-loaded molds were placed within the melt press and compressed. A vacuum was drawn down to a few thousand Pascals, and the heat was turned on. During the heating and cooling cycles, the samples were further compressed every 10 min to drive air bubbles out. Once the melt press reached the desired temperature of 150 °C (for instance), the heat was turned off and the assembly was allowed to cool to 50 °C at which time the mold was removed. The fully formed samples were then separated from the mold. An image of the initial pellets and fully formed sample disks is shown in Fig. 1. It should be noted that due to this processing step and previous drying steps, it is unlikely that any treated samples will have significant residual solvent. The final sizes of the shear rheology disks were 25 mm in diameter and between 1.2 and 1.3mm in thickness. The cylinders used for extensional rheology had a diameter of 10 mm and a height ranging between 6 and 7 mm.

Differential scanning calorimetry (DSC)

Differential scanning calorimetry (DSC) was performed on each of the virgin and toluene-treated sample pellets using a TA Instruments Discovery D25 with RCS 90. Each test involved 4–5 mg of sample placed within a hermetically sealed aluminum pan. Samples were heated to 175 °C, cooled to 75 °C, and heated a second time to 175 °C. Each temperature sweep utilized a ramp speed of 10 K/min while in a nitrogen environment with a volume flow rate of 50.0 mL/min. TRIOS analysis software from TA instruments was used to determine the peak minimum and maximum for the melt and crystallization temperatures, respectively. The degree of crystallinity is also of interest and can be defined as

Fig. 1 Initial appearance of the LDPE pellets and the discs vacuum pressed for the shear rheometer

$$DoC = \Delta H/\Delta H_u \quad (1)$$

where ΔH is the latent heat of fusion obtained by integrating under the DSC curves and $\Delta H_u = 293 J/g$ is the latent heat of fusion for PE per unit mass assuming 100% crystallization taken from the literature (Chang 1974).

Shear rheology

Small amplitude oscillatory shear (SAOS) and steady shear measurements were carried out on a Kinexus Pro+ rheometer from Malvern Instruments, employing a 25-mm-diameter parallel-plate geometry. All analyses of the shear rheology results were carried out using a commercially available rheology software tool (IRIS) (Poh et al. 2022). All experiments were performed in a nitrogen environment to minimize the oxidation and enhance the polymer's stability over time. However, even under nitrogen protection, degradation and cross-linking of the molten polymer will eventually occur. Identifying the specific temperatures and times at temperature where each of the polymers remained stable was critical for both shear and extensional rheology measurements. The stability of the polymer was evaluated using time-resolved mechanical spectroscopy (TRMS), both in air and under nitrogen (Mours and Winter 1994; Poh et al. 2022).

As proposed by Mours and Winter, due to the high sensitivity of rheology to small changes in polymer architecture and composition, time-resolved mechanical spectroscopy (TRMS) is a very effective tool for assessing polymer stability and tracking sample alterations during processes like crosslinking, gelation, or degradation (Mours and Winter 1994). As depicted in Fig. 2a, TRMS involves conducting successive frequency sweeps to elucidate the evolution of material properties such as the storage modulus, G', loss modulus, G'', and complex modulus, G^*, over time. In Fig. 2b, we present measurements of the fluid elasticity,

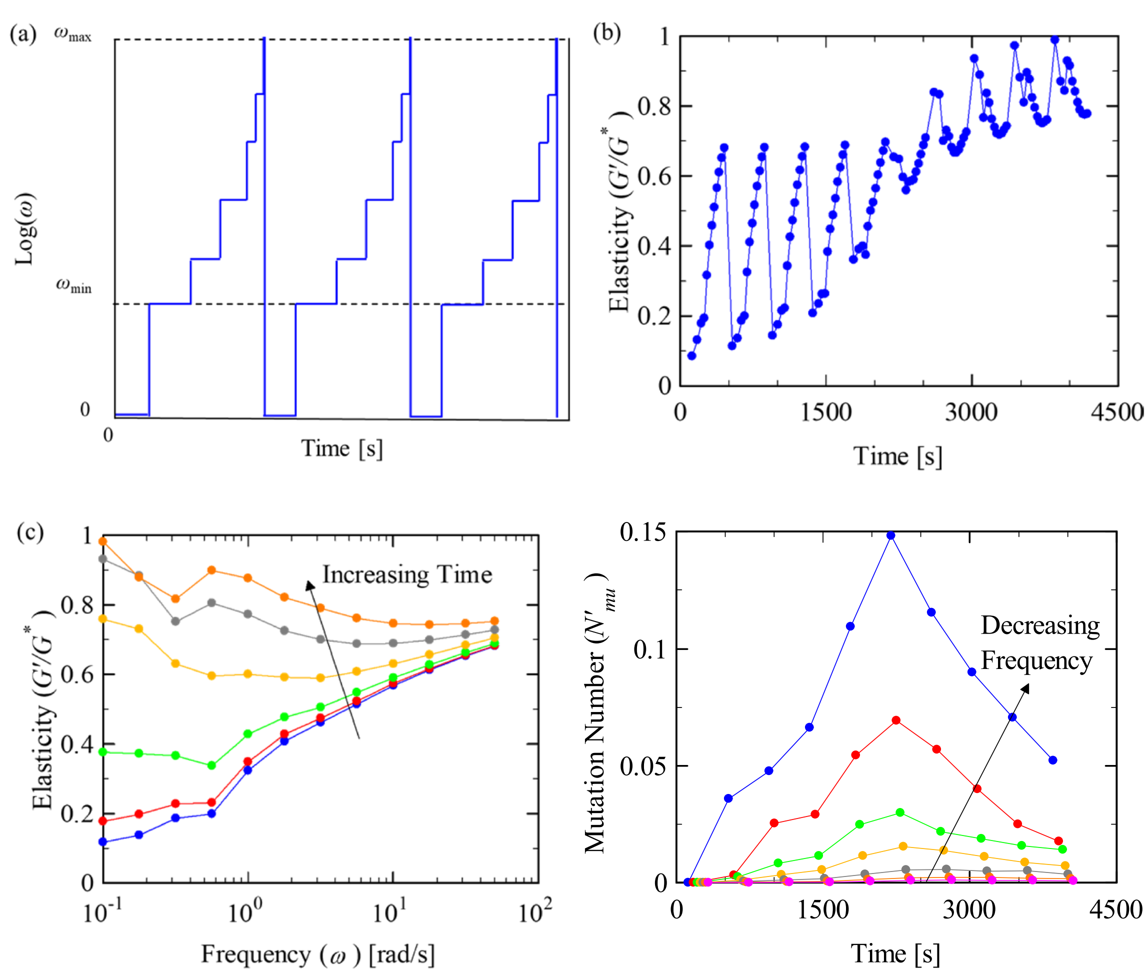

Fig. 2 Time resolved rheology of a virgin LDPE taken at 200 °C. **a** Inputted frequency from the instrument over time, for visual clarity only 5 frequencies are shown, data is taken over 12 frequencies, **b** the corresponding elasticity results over time, **c** elasticity results instead versus frequency for times of 10 to 60 min, and **d** interpolated results using TRMS showing mutation number vs. time for frequencies of 0.1 to 50 s^{-1}

G'/G^*, of the virgin LDPE at 200 °C as a function of time to illustrate the usefulness of this TRMS technique. If the polymer melt were stable, rheological properties like fluid elasticity would repeat the same path over and over again without changes. However, we deliberately chose a sample that clearly crosslinks, gels, or degrades with time. In Fig. 2b, the elasticity remains stable for roughly 1000 s before beginning to increase with time. In general, it is more informative to view the development of material properties versus frequency as a typical SAOS experiment would be presented. To do this, using the IRIS Rheo-Hub software, each of the single-frequency data set is fit with a smoothed curve and interpolation of those curves is performed to produce a series of G' and G'' versus ω data sets at different experimental times as shown in Fig. 2c (Mours and Winter 1994). Generally, a more informative approach involves visualizing material property development against frequency, similar to a conventional small-amplitude oscillatory shear (SAOS) experiment. These representations may encompass the elasticity that is shown in Fig. 2c or they could be the storage or loss moduli, the loss angle, the dynamic viscosity, or even the relaxation time spectrum. The data in the "Results" section that follow will focus solely on the processed data like that in Fig. 2c.

TRMS can be applied as long as the material changes are sufficiently slow. In other words, the material must behave in a quasi-stable manner for the data to be viable (Mours and Winter 1994). If the material changes too quickly, the material response will become non-linear and the TRMS data becomes unreliable. Quasi-stability of the material can be determined by verifying the mutation number, $N_{mu} = \Delta t/\lambda_{mu}$, is sufficiently small. The mutation number relates the characteristic time for the material changes also called the mutation time, $\lambda_{mu} = |\delta g/g\delta t|$, and the experimental time, Δt. For TRMS, the experimental time can be approximated as the period of the oscillatory shear, $\Delta t \approx 2\pi/\omega$. The mutation number typically needs to be evaluated for both G' and G''; however, for transient materials G' responds more sensitively; N'_{mu} will be of greater importance than N''_{mu}. The mutation number based on G' is as follows:

$$N'_{mu} = \frac{2\pi}{\omega G'}\left|\frac{dG'}{dt}\right| \tag{2}$$

For mutation number less than $N'_{\mathrm{mu}} < 0.1$, the samples can be considered quasi-stable during the measurement of each individual data point and the TRMS data can be considered valid (Mours and Winter 1994). A plot of N'_{mu} for the virgin LDPE at 200 °C can be seen in Fig. 2d. Here the mutation number for the higher frequencies (0.1778 to 50 s^{-1}) does not exceed 0.1 and does not need to be excluded. The mutation number at the lowest frequency (0.1 s^{-1} in blue) exceeds 0.1, suggesting that those data are invalid and need to be excluded from the rheology presented in Fig. 2c. In future TRMS measurements, only data for which $N'_{\mathrm{mu}} < 0.1$ will be presented.

Once the stability window had been identified, SAOS measurements were conducted. Frequency sweeps between $0.1 < \omega < 60$ rad/s were performed to determine the storage and loss modulus of each polymer at a given temperature while maintaining the strain amplitude within the linear viscoelastic regime. In this case, a percent strain of 7% was used for each of the samples tested. The lower end of the frequency range was chosen to reduce the time at temperature needed for each frequency sweep to minimize the likelihood of polymer degradation and large mutation numbers. The frequency data can be further processed using time-temperature superposition (TTS) to expand the frequency range and to produce a relaxation time spectrum; however, the shift factors for PE are known to be quite small and given the concerns of polymer degradation at high temperatures, TTS was not performed here. The moduli are very sensitive to small changes in polymer molecular weight distribution, branching, and cross linking, all of which can change during the recycling processes.

Steady shear measurements were also performed for both the virgin and toluene-treated HDPE and LDPE samples. The master curves were then fit with a Carreau model (Yasuda et al. 1981) to determine the characteristic material times and the shear thinning power law index.

Filament stretching extensional rheology (FiSER)

A UMass-built filament stretching extensional rheometer (FiSER) was used to apply a homogeneous uniaxial extensional flow to both the virgin and toluene-treated LDPE and HDPE samples (Rothstein and McKinley 2002a, 2002b; Rothstein 2003; Chellamuthu et al. 2011; White et al. 2012). A schematic of the FiSER and an oven designed for stretching polymer melts are shown in Fig. 3. The FiSER oven is designed to enclose the fluid filament as it is stretched. The temperature can be controlled up to 250 °C at an accuracy of ± 1 K throughout the oven. This is achieved using four independently controlled heating zones separated by horizontal plates to reduce natural convection and an additional localized heating zone in close proximity to the bottom endplate. A complete description of the operating space and the design of the FiSER and the oven used here can be found in the literature (Rothstein and McKinley 2002a; 2002b; Rothstein 2003; Chellamuthu et al. 2011; White et al. 2012). It should be noted that the FiSER involves too large of a heating chamber to allow for nitrogen protection. Prior to testing, TRMS measurements were performed without nitrogen to verify a stability window.

In a FiSER experiment, a cylindrical sample is placed between two endplates and stretched while the force and

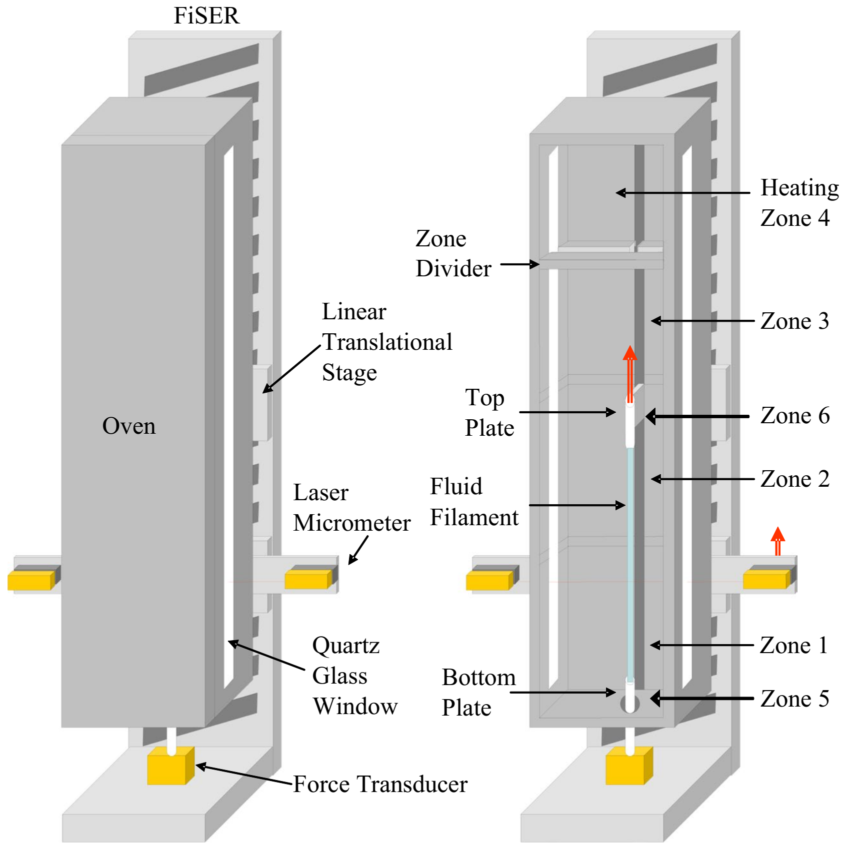

Fig. 3 Schematic diagram of a filament stretching rheometer with custom built oven

midpoint radius are measured simultaneously. Midpoint radius readings are fitted using a third-order polynomial to reduce noise in calculations that require differentiation of the data. The goal of FiSER is to impose a motion on the endplates seen in Fig. 3, such that the extension rate on the fluid defined as the following is constant:

$$\dot{\varepsilon} = -\frac{2}{R_{mid}(t)}\frac{dR_{mid}(t)}{dt} \tag{3}$$

Here R_{mid} is the radius at the midplane of the stretching fluid filament measured by a laser micrometer programmed to follow along at half the speed of the upper endplate at the midway point between the two end plates. An open-control loop is used to impose a constant extension rate. Closed-loop control is possible on our FiSER, but not for extension rates beyond 0.2 s^{-1} due to the limitations of the data acquisition rate of the instrument. At the lower strain rates, the data was limited at large strains due to the resolution limit of the 100-g force transducer.

The strength of the extensional flow is defined by the Weissenberg number, $Wi = \lambda\dot{\varepsilon}$, while the deformation of the polymer melt is described by the Hencky strain, $\varepsilon = -2\ln(R_{mid}/R_0)$. The FiSER used in these experiments can obtain an extension rate of up to $\dot{\varepsilon}_{\max} = 10.0$ s^{-1} and apply enough deformation to obtain a Hencky strain up to approximately $\varepsilon_{max} = 10$. The elastic tensile stress difference generated within the filament, $\langle\tau_{zz} - \tau_{rr}\rangle$, is calculated from the force measured at the load cell while taking into account the inertia of the fluid, the surface tension of the fluid, and the fluid's weight (Szabo 1997; Anna et al. 2001). The extensional viscosity can be calculated by dividing the elastic tensile stress by the extension rate imposed on the fluid: $\eta_E = \left\langle\tau_{zz} - \tau_{rr}\right\rangle/\dot{\varepsilon}$. The extensional viscosity can then be non-dimensionalized by the zero-shear-rate viscosity to create a Trouton ratio, $Tr = \eta_E/\eta$. Here the shear viscosity in the Trouton ratio is evaluated at a shear rate equal to the applied extension rate. The extensional viscosity can be plotted alongside the shear viscosity to see the effect of fluid elasticity. For a Newtonian fluid, the Trouton ratio is exactly $Tr = 3$. However, for a polymer solutions and melts, when the Weissenberg number is greater than one-half, $Wi > ½$, polymer elongation during stretch can lead to strain hardening and very large Trouton ratios, $Tr >> 3$ (Münstedt and Laun 1981; Mckinley and Sridhar 2002).

The extensional viscosity is highly sensitive to changes in molecular weight, molecular weight distribution, polymer architecture, and cross linking. Extensional rheology measurements have been effectively used in the past to observe changes to polymer melts that can occur at high temperature and which could also occur during recycling processes (Sur et al. 2019, 2020).

Results and discussion

Degree of crystallinity

Differential scanning calorimetry (DSC) was performed to determine the impact of solvent treatment on the melt temperature and degree of crystallinity of both the LDPE and HDPE samples. Figure 4 presents a typical DSC result comparing the virgin and toluene-treated HDPE samples. Table 1 summarizes the important data for both the virgin and toluene-treated LDPE and HDPE. Samples underwent two heating/cooling loops of 10 K/min during each DSC measurement. The initial heating step removed the previous heat and deformation history of the sample. The sample was then cooled to identify the crystallization temperature and reheated to determine the melt temperature and degree of crystallinity. The appropriate temperature for sample preparation and rheological testing was determined based on the DSC's nominal melt temperature.

In Fig. 4, during the first heating cycle, differences between the virgin and toluene-treated HDPE samples are evident. During the first heating cycle, the melt temperature of the virgin HDPE was on average 3.5 K higher than in the second heating cycle across three DSC measurements. However, the toluene-treated HDPE showed little difference in melt temperature between the first and second heating cycles. This difference likely stems from residual effects of the thermal history that the virgin HDPE experienced when it was initially formed into pellets. The toluene-treated HDPE was not formed into pellets, but provided as flakes that precipitated from the toluene solution. After the thermal history effects of the virgin HDPE were eliminated in the first heating cycle, only marginal differences of melt and crystallization temperatures between the virgin and toluene-treated samples remained, with an average of 0.5 K attributed to toluene treatment for the second heating cycle across three DSC measurements. As shown in Table 1, the HDPE has a nominal melt temperature of T_m = 130 °C melt and a nominal crystallization temperature of T_c = 111 °C. Based on this information, it was concluded that sample preparation and rheological characterization for the HDPE should be carried out at or above 140 °C.

The large change in melt temperatures between the first and second heating ramp that was seen for the virgin HDPE did not occur for the LDPE. The LDPE was found to have a nominal melt temperature of T_m = 113 °C and a nominal crystallization temperature of T_c = 96 °C, as shown in Table 1. Based on this data, it was determined that sample preparation and rheological characterization for the LDPE should be conducted at or above 120 °C.

The area under the DSC curves was integrated in order to calculate the degree of crystallization for each sample from

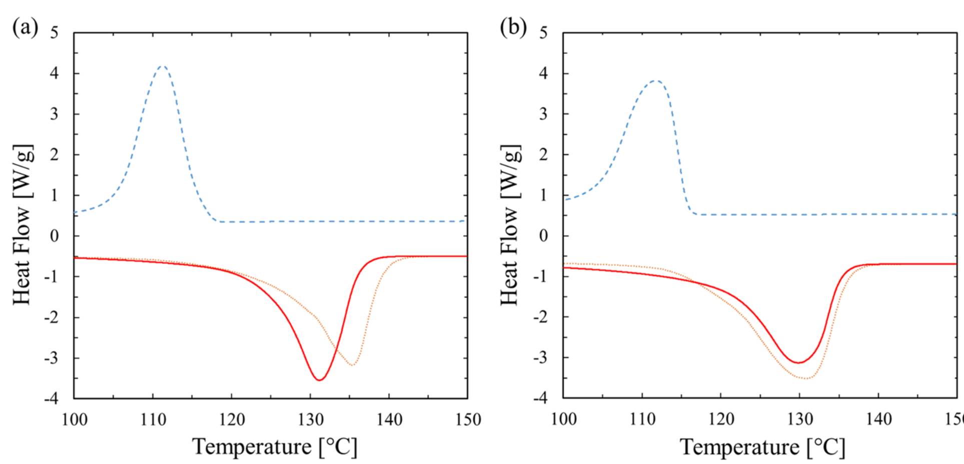

Fig. 4 DSC using 10 K/min ramps for the HDPE: **a** virgin and **b** toluene treated. The dotted orange represents the initial heating, dashed blue for the cooling, and solid red for the second heating

Table 1 DSC data for HDPE and LDPE using a 10 K/min temperature ramp. Data are averaged from three different DSC measurements for each of the four sample types

Sample name	Second ramp melt temp (°C)	Crystallization temp (°C)	Degree of crystallinity first heat ramp (%)	Degree of crystallinity second heat ramp (%)	Degree of crystallinity cooling ramp (%)
LDPE	114	96	22	19	22
Treated LDPE	113	96	36	18	21
HDPE	131	111	52	52	49
Treated HDPE	130	112	67	49	47

Eq. (1). As shown in Table 1, the degree of crystallization tabulated for the virgin LDPE and HDPE samples did not change significantly between heating cycles. For the virgin LDPE, the degree of crystallization was found to be DoC = 22%, while for virgin HDPE it was 52%. These values are typical for these polymers. However, the toluene treatment had a notable impact on the degree of crystallinity for both the LDPE and HDPE samples. The DoC of the toluene-treated LDPE decreased from 36% crystallinity to 18% from the first to the second heating cycle. A similarly large decrease was observed for the DoC of the toluene-treated HDPE which decreased from 67 to 49% from the first to the second heating cycle. This large drop in crystallinity likely stems from the difference in how the polymers were processed. For the virgin samples, the polymer crystallized as it was cooled from its molten state following being formed into pellets. The toluene-treated samples, however, were formed as they precipitated out of solution. While in the toluene, the polymer is swollen and more mobile, allowing the polymer chain segments to more readily sample the space around them, more quickly find neighboring chains, reorient, and likely crystallize more easily (Gao et al. 2012). Additionally, it is possible that not all of the crystallized polymer fully dissolved in toluene, leaving undissolved crystals that could act as nucleation sites, increasing the polymer crystallinity as it precipitated out of solution (Camacho et al. 2017; Sturm et al. 2018).

Polymer stability

Compared to industrial processing times, the residence time for rheological testing can be much longer. Even at relatively low temperatures, polymer samples are prone to structural changes, such as polymer degradation and cross-linking, even when in a nitrogen environment. To ensure the stability of polymer samples and the usefulness of rheological data, it is essential that the samples do not undergo significant changes over the course of the testing. For this reason, before conducting any shear or extensional rheology measurements, the first round of rheological testing addressed the polymer stability using the TRMS framework outlined in the "Shear rheology" section. Using TRMS, the variation in storage and loss modulus was captured over a wide range of angular frequencies for temperatures between 150 and 200 °C. Figure 5 summarizes the stability results for the LDPE (Fig. 5a) and HDPE (Fig. 5b) by displaying the variation of the fluid elasticity, G'/G^*, over time at an applied angular frequency of ω = 0.1778 rad/s at several different temperatures. Although not shown here, similar ranges of stability were observed for the LDPE and HDPE in both nitrogen and air.

From the data in Fig. 5, it is evident that virgin LDPE at 200 °C is unstable and that the rheological measurements cannot be confidently carried out at this temperature. The elasticity increases very quickly from 0.13 to 0.89 over the course of 1 h of testing, indicating that crosslinking, gelation, or degradation is occurring. However, by reducing the temperature of the virgin LDPE from 200 to 175 °C, it became stable throughout the entire hour of testing. Here, less than a 4% change was observed. The toluene-treated LDPE at 175 °C was found to be slightly less stable than the virgin LDPE at the same temperature, with a 10% increase in elasticity after 50 min. Moving forward, a maximum of 10% variation in elasticity was chosen as the limit for stability in rheological testing. As a result, all rheological measurements for the toluene-treated LDPE presented here will be performed at 175 °C with test times less than 50 min.

Similar TRMS studies on HDPE are shown in Fig. 5b. The HDPE was found to be less stable than the LDPE. In fact, TRMS data for the virgin HDPE at 200 °C are not shown because the mutation number exceeded the acceptable limits, $N_{mu} > 0.1$. The toluene-treated sample was found to be unstable even at 175 °C. However, the virgin HDPE at 150 °C was found to be stable over the entire hour-long test, with the elasticity remaining within the 10% variation limit. The toluene-treated HDPE at 150 °C was stable for approximately 30 min before the elasticity increased by 10%. All experiments for HDPE will thus be performed at 150 °C with test times less than 30 min.

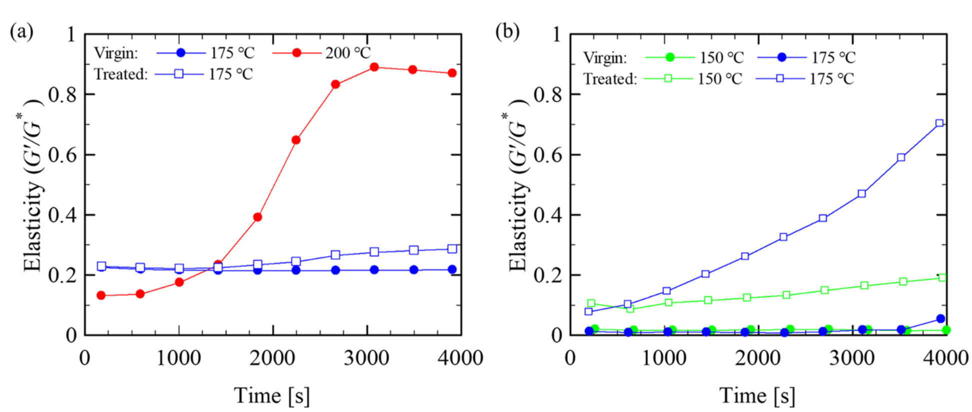

Fig. 5 TRMS stability analysis depicts fluid elasticity over time for **a** LDPE and **b** HDPE at different temperatures. All data are for an applied oscillatory frequency of ω = 0.1778 rad/s

In addition to the fluid elasticity, changes to the fluid complex viscosity, η^*, over time can also be a strong indicator of the fluid stability as shown in Fig. 6. The data show that temperature-induced changes to the polymer primarily affect viscosity data primarily at the low frequencies. For example, at the lowest frequency of 0.1 rad/s, viscosity of the virgin LDPE presented in Fig. 6a was found to decrease by 7% from 4.6 to 4.3 kPa-s, while the viscosity of toluene-treated LDPE was found to increase by 15% from 5.2 to 6 kPa s. The trends for HDPE are even more pronounced. At 0.1 rad/s, the complex viscosity of virgin HDPE dropped by 2% from 0.92 to 0.90 kPa-s, while toluene-treated HDPE increased by 42% going from 2.2 to 3.1 kPa-s. These observations suggest that one of the main effects of the toluene treatment is to destabilize the polymer melts by removing toluene-soluble stabilizers from the melts during solvent treatment. These stabilizers are purposefully added during the manufacturing of the virgin polymers.

Furthermore, the toluene-treated LDPE and HDPE exhibited higher overall viscosity when compared against the virgin samples of each. For the LDPE, the complex viscosity increased by as much as 30% while for the HDPE, the increase in viscosity after toluene treatment was much larger with increases of more than 250% measured at the lowest frequencies tested. These changes are not the result of polymer stability and thus they must result directly from the toluene treatment of the sample. It is our hypothesis that the solvent wash process stripped out not just the polymer stabilizers but also the low molecular weight components of the polymer melts that can act as plasticizers in the polymer melt, reducing the viscosity and relaxation time of the polymer melt. These low molecular weight tails of the polymer molecular weight distribution can remained soluble in toluene even as the temperature was dropped and the higher molecular weight components precipitated out of solution. In order to test this theory, gel permeation chromatography (GPC) of the PEs is included in Figs. S1 and S2 for molecular weights between 10^3 and 10^7 g/mol. A negligible difference in the distribution is seen as a result of toluene treatment for either the LDPE or the HDPE samples over this range of molecular weights. However, significant differences were observed for much lower molecular weight PEs. Gas chromatography–mass spectrometry (GC-MS) was performed on the toluene filtered from the solution after the polymer precipitated out to determine the fraction of molecular weight components that are below the 10^3 g/mol lower limit of the GPC data.

Solvent wash contents

The toluene samples recovered after each polymer dissolution were analyzed with GC-MS to identify any unknown compounds in the solvent that were removed from the virgin polymers. These results are shown in Fig. 7 for the different samples; more detailed results can be seen in Fig. S3 of the Supplemental Information. For the LDPE, in addition to toluene and its impurities, three major compounds were detected in the toluene after LDPE dissolution. Short-chain PE oligomers, in the range of C9 to C35, remained in the solvent with concentrations from 5 to 32 PPM when the ratio of solvent to polymer is 1:5. Removal of these short-chain oligomers caused an increase of the number-average molecular weight (M_N) and a decrease in the dispersity index (PD) of the LDPE after STRAP, suggesting a narrower molecular weight distribution. In addition to this, a plasticizer, diethylene glycol dibenzoate, was also detected in the solvent at a concentration of 16.8 PPM. Plasticizers are common resin additives that aid in the processability of different polymers, lowering the viscosity, and enhancing the overall stability (Wypych 2017). Removal of the plasticizer from the LDPE would cause an increase in its viscosity, which was observed in the TRMS stability analyses for complex viscosity which was discussed in the "Polymer stability" section. Lastly, fluorinated compounds were detected in the toluene after LDPE dissolution, most likely per- or polyfluoroalkyl substances (PFAS) that would be present in the virgin polymer to provide stability. Our GC-MS method was able to identify the presence of PFAS but could not discern

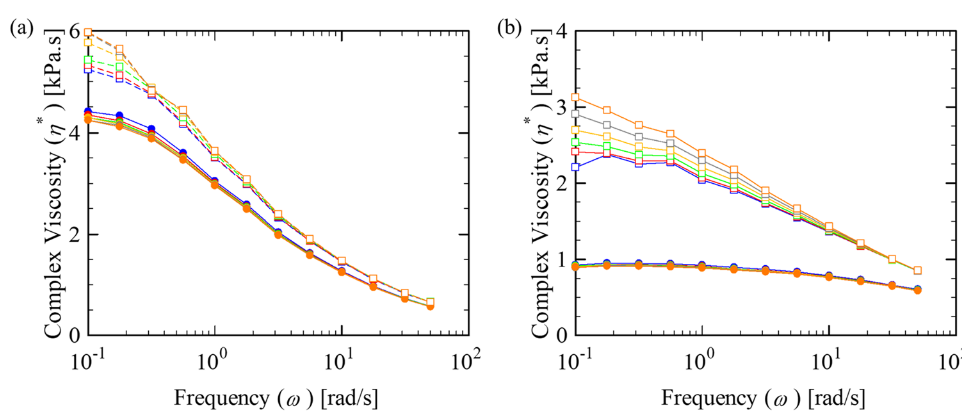

Fig. 6 TRMS stability analysis showing complex viscosity as a function of frequency for **a** LDPE at 175 °C and **b** HDPE at 150 °C. Data is presented in increments of 10 min from 10 to 60 min, blue 10 min, red 20, green 30, yellow 40, gray 50, and finally, orange 60 min

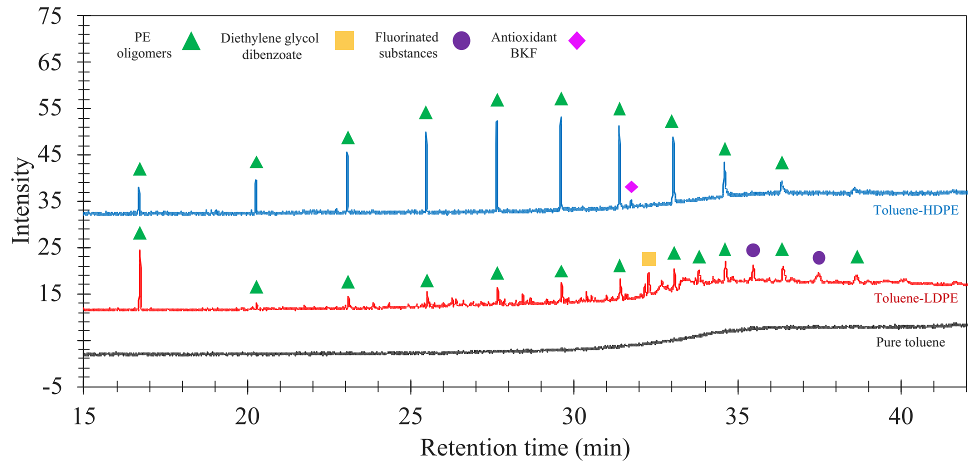

Fig. 7 Chromatogram comparison of pure toluene (in black), toluene after LDPE dissolution (in red), and toluene after HDPE dissolution (in blue). GC-MS method: DB-5 column, helium carrier gas at 8.7 mL/min, and a maximum oven temperature of 310 °C

between specific structures or its origin. Removal of PFAS likely decreased the stability of the toluene-treated LDPE causing it to degrade at lower temperatures than the virgin LDPE. Similarly, the toluene sample after HDPE dissolution also showed the presence of short-chain PE oligomers, with the possibility of having chains longer than C40. Specifically in the range of C16 to C40, concentrations in the solvent after HDPE dissolution were from 6 to 110 PPM. The diethylene glycol dibenzoate or the fluorinated substances were not detected in the toluene after HDPE dissolution; however, an antioxidant BKF was observed in the solvent. The presence of plastic additives in the toluene will depend on the affinity of the plastic additive to stay in solution and the initial composition of the resins that are considered for the dissolution process. Additional methods will be required to characterize non-volatile components and have a complete analysis on additive accumulation in the solvents.

Relaxation time spectrum

Using the stability results from the "Polymer stability" section, we identified a temperature and time at temperature where each sample remained stable, ensuring confidence in the rheological data presented here and in the next section. For the LDPE, a temperature of T = 175 °C was chosen, while for the HDPE a temperature of 150 °C was used. A nitrogen bath was utilized for all DSC and shear rheology experiments. For the small amplitude oscillatory shear (SAOS), an amplitude sweep on the virgin LDPE and HDPE was initially carried out at an angular frequency of $\omega = 1$ s^{-1} to determine the upper bounds of the strain that could be applied while remaining within the linear viscoelastic regime. The virgin LDPE remained in the linear viscoelastic regime up to 16% strain at 175 °C, while the virgin HDPE remained in the linear viscoelastic regime up to 32% strain at 150 °C. A strain amplitude of 7% was chosen to ensure that the SAOS would remain within the linear range. In Figs. 8 and 9, SAOS and steady-shear experiments of the virgin and toluene-treated samples are presented. The storage, G', and loss modulus, G'', for both the virgin and the toluene-treated samples in Figs. 8a and 9a are typical of polydisperse polymer melts. Due to stability issues, obtaining lower frequency data was difficult due to the long times required for the measurements. Typically, time-temperature-superposition can be used to extend the data to lower frequencies, but again thermal stability limited the temperature range over

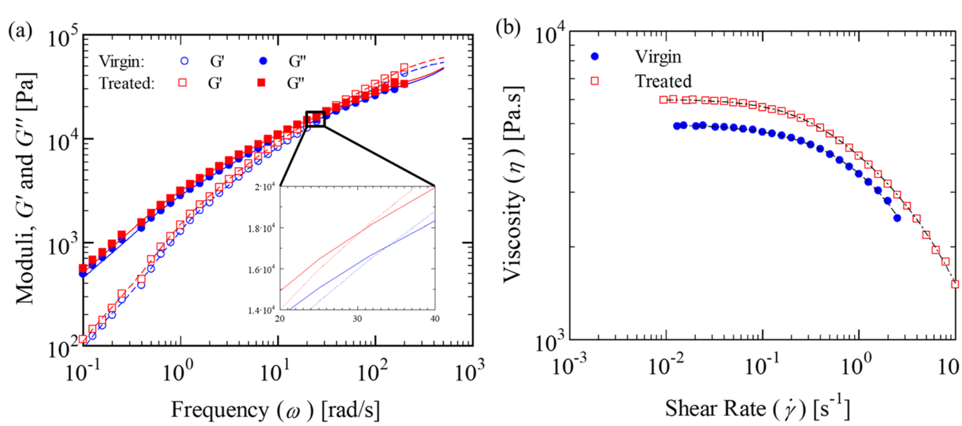

Fig. 8 Shear rheology of the LDPE at 175 °C: blue circles virgin and red squares for toluene treated. **a** Frequency sweep with multi-mode Maxwell model (Baumgaertel and Winter 1989, 1992) superimposed, G' with filled symbols, G'' with open symbols; **b** steady shear with a Carreau model (Yasuda et al. 1981) superimposed

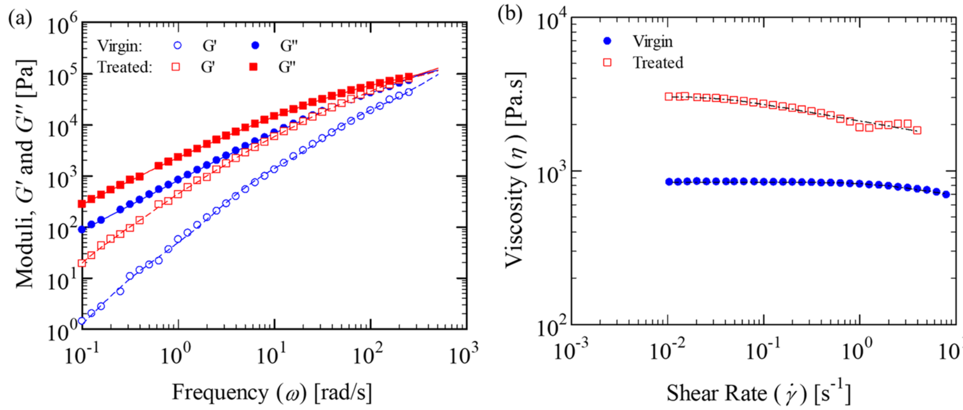

Fig. 9 Shear rheology of the HDPE at 150 °C: blue circles virgin and red squares toluene treated. **a** Frequency sweep with multi-mode Maxwell model (Baumgaertel and Winter 1989, 1992) superimposed, G' with filled symbols, G'' with open symbols; **b** steady shear with a Carreau model (Yasuda et al. 1981) superimposed

which measurements could be made. As a result, the data in Figs. 8a and 9a do not fully enter the terminal regime although data is obtained for nearly four orders of magnitude in angular frequency.

Analyzing the SAOS data for the LDPE, it is not easy to discern the differences between the moduli of the virgin and the toluene-treated LDPE from the log-log plot in Fig. 8a. However, subtle distinctions emerge when the data are plotted on linear axes and zoomed into a small region of the data where the storage and loss moduli G' and G'' cross, as is done in the inset in Fig. 8a. Both the storage and loss modulus of the toluene-treated samples consistently exceed those of the virgin sample by approximately 15%. This shift in moduli also affects the relaxation time spectrum. The cross-over point of the G' and G'' is indicative of the fluid relaxation time, λ. It is evident from Fig. 8 that toluene-treated LDPE has a larger relaxation time than the virgin sample. The cross-over for the toluene-treated LDPE is at a frequency of $\omega = 30$ rad/s, while for the virgin sample it is at $\omega = 34$ rad/s. Note that a representative relaxation time can be defined as the inverse of the cross over frequency, $\lambda = 1/\omega$, which would suggest a 10% increase in relaxation time with toluene treatment. A more detailed measure of the relaxation phenomenon requires determination of the entire relaxation time spectrum. A commercially available rheology software tool (IRIS) was used to fit the data with multi-mode Maxwell model to determine the relaxation time spectrum (Baumgaertel and Winter 1989, 1992). These fits are superimposed over the data in Fig. 8a and are presented in Table 2. The individual discrete modes can be seen in Figs. S1 and S2 of the supplemental information. The relaxation modulus $G(t)$ can be represented by a sum of discrete Maxwell modes with partial relaxation moduli g_i and relaxation times τ_i:

$$G(t) = \sum_{j=1}^{n} G_i(t) = \sum_{j=1}^{n} g_j \exp\left(-t/\tau_j\right) \tag{4}$$

For both the LDPE and HDPE, toluene treatment caused an increase in the sum of discrete modes. The slowest relaxation process defines the reptation or disengagement time, λ_d, that is of primary importance in extensional rheology measurements, because in order for the tube of constraints to become aligned in extensional flows, a Weissenberg number of $Wi = \lambda_d \dot{\varepsilon} > 0.5$ is needed (Feldman 1989). The disentanglement time was taken to be the viscosity-weighted-average relaxation time of the n-mode fit:

$$\lambda_d = \sum_{i=1}^{n} \eta_i \lambda_i / \sum_{i=1}^{n} \eta_i \tag{5}$$

Results of these calculations are in Table 2 for the virgin and the toluene-treated LDPE samples. As can be seen, an 11% increase in the disentanglement time was observed for the toluene-treated LDPE.

Table 2 Multi-mode Maxwell modeling (Baumgaertel and Winter 1989, 1992) of the frequency sweep trials, from Figs. 8a and 9a. Carreau modeling (Yasuda et al. 1981) constants of steady shear trials, from Figs. 8b and 9b. The virgin and treated HDPEs were tested in triplicate and averaged

Model	Multi-mode Maxwell		Carreau		
	Sum of discrete modes G [MPa]	Disentanglement time λ_d [s]	Zero shear viscosity η_0 [kPa·s]	Characteristic time for shear thinning k [s]	Power law index n
Virgin LDPE	0.63	2.7	4.9	3.4	0.15
Treated LDPE	0.57	3.0	5.8	2.6	0.19
Virgin HDPE	7.7	0.40	0.87	0.88	0.044
Treated HDPE	4.8	2.3	2.8	18	0.071

For the toluene-treated HDPE because of the instrument noise and large instrument inertia at these high frequency, precise determination of the cross over point is difficult. Instead, a more accurate comparison can be made by focusing on the calculations of the disentanglement time shown in Table 2. Here the increase in relaxation time for the treated HDPE is found to increase from $\lambda_d = 0.40$ s for the virgin HDPE to $\lambda_d = 2.3$ s for the toluene-treated HDPE. This factor of 5.8× change in the relaxation time will have a significant effect on the processability and end uses of the recycled HDPE.

Changes were also observed with toluene treatment for the steady shear viscosity in Figs. 8b and 9b. Here a 20% increase in the zero-shear-rate viscosity, η_0, accompanied by an early onset of shear thinning was observed after toluene treatment of the LDPE. For the HDPE, over a 3× increase in the shear viscosity was observed. The HDPE data also shows a much early onset of shear thinning consistent with the large increase in the relaxation time. The data was fit with the Carreau model to quantify the degree of shear thinning (Yasuda et al. 1981):

$$\eta(\dot{\gamma}) = \eta_0\left[1 + (k\dot{\gamma})^2\right]^n \qquad (6)$$

Here k is the characteristic time and n is the power law index that describes the strength of shear thinning. These fits to the Carreau model are superimposed over the data in Figs. 8b and 9b and are presented in Table 2. For both the LDPE and the HDPE, toluene treatment was found to increase the zero-shear-rate viscosity and marginally increase the strength of the shear thinning. Toluene treatment decreased the characteristic time of the LDPE and increased the characteristic time for the HDPE.

The increase in the toluene-treated sample moduli, viscosity, and relaxation time can be attributed to the removal of low MW fractions and plasticizer as demonstrated by the "Solvent wash contents" section. The removal of stabilizer could also affect these parameters by means of cross-linking or growth of side chains or branches during rheological testing, however, the stability testing in the "Polymer stability" section indicates that it is unlikely. In Fig. 2b the LDPE at 200 °C just past 1000–1500 s there is a drastic increase in the elasticity, it is likely this instability is crosslinking occurring. For crosslinking to be occurring in Figs. 8a and 9a, we would expect to see a plateau or at least an increase in the low-frequency data that is not present. Additionally, these factors would result in overall shift in the molecular weight distribution to a higher value, which is not supported by the GC-MS or GPC data sets.

Strain hardening

A series of transient uniaxial extension measurements were performed on the virgin and toluene-treated LDPE and HDPE. This is illustrated in Fig. 10 for LDPE and Fig. 11 for the HDPE for stretching over a range of extension rates between 0.2 s^{-1} < $\dot{\varepsilon}$ < 2.0 s^{-1}. The linear viscoelastic response of each fluid is superimposed over the data and shows the expected good agreement with the extensional rheology data in the low strain, low strain-rate limit. For Weissenberg number of the stretch greater than $Wi = \lambda_d\dot{\varepsilon} >$ 0.5, a viscoelastic response that includes strain hardening of the extensional viscosity is expected. Based on the measurements of the disentanglement times made in shear and presented in Table 2, strain hardening of the extensional viscosity is expected above an extension rate of roughly $\dot{\varepsilon} > 0.3$ s^{-1} for all but the toluene-treated HDPE. For that case, due to its significantly smaller relaxation time, a viscoelastic response is only expected above $\dot{\varepsilon} > 2.5$ s^{-1}. As seen in Figs. 10 and 11, some degree of strain hardening was observed for all the samples for $Wi > 0.5$, although the strain hardening of the virgin HDPE was found to be quite weak even at the largest extension rates tested. The general trends in the extensional rheology response of the virgin LDPE and HDPE are consistent with the literature (Münstedt and Laun 1981; Huang et al. 2016; Morelly and Alvarez 2020).

For the both the virgin LDPE and HDPE samples, at the lowest extension rates tested, no strain hardening was observed and the extensional viscosity was found to be roughly three times the shear viscosity in each case resulting

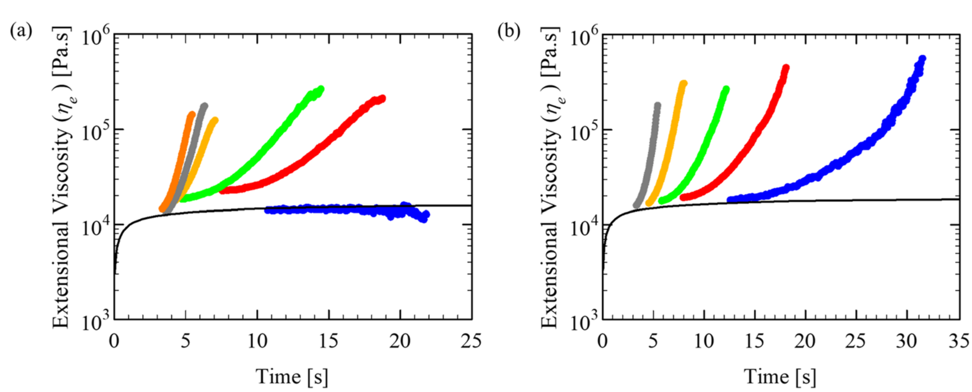

Fig. 10 Uniaxial extensional rheology of the LDPE at 175 °C without nitrogen: **a** virgin and **b** toluene treated. In blue is an extension rate of 0.2 s^{-1}, red 0.4 s^{-1}, green 0.6 s^{-1}, yellow, 1 s^{-1}, gray 1.4 s^{-1}, and orange 2 s^{-1}. The linear viscoelastic envelopes calculated from Fig. 8a are included and shown in black

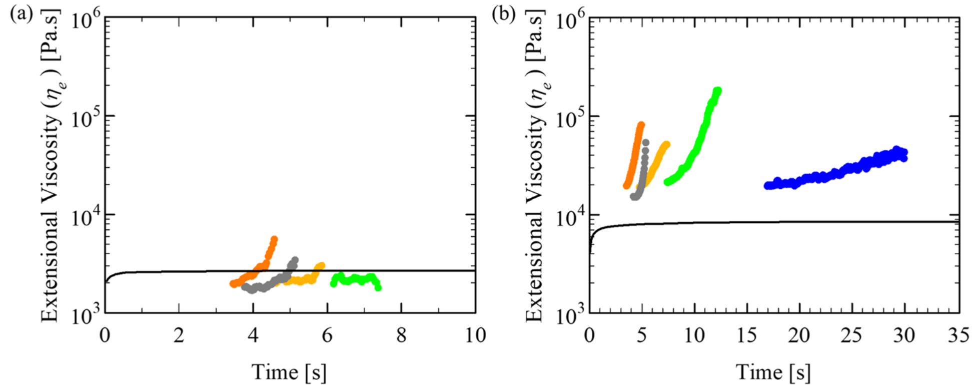

Fig. 11 Uniaxial extensional rheology of the HDPE at 150 °C without nitrogen: **a** virgin and **b** toluene treated. In blue is an extension rate of 0.2 s^{-1}, green 0.6 s^{-1}, yellow, 1 s^{-1}, gray 1.4 s^{-1}, and orange 2 s^{-1}. The linear viscoelastic envelopes calculated from Fig. 9a are included and shown in black

in a Trouton ratio of roughly $Tr = \eta_e/\eta \approx 3$. For the toluene-treated LDPE and HDPE, strain hardening was observed even at the lowest extension rates tested due to the increase in relaxation time observed in the shear rheology following toluene treatment. When comparing the LDPEs and HDPEs, it is evident that the HDPE strain hardens significantly less; this is due to less branching being present in the HDPE samples (Morelly and Alvarez 2020; Poh et al. 2022). At large Weissenberg numbers and Hencky strains, $\varepsilon = \dot{\varepsilon}t$, deformation and stretching of the polymers led to strain hardening of the extensional viscosity and the resulting Trouton ratios were found to grow quite large for all but the virgin HDPE sample. For all four samples tested, with increasing values of imposed extension rate and Weissenberg number, the onset of strain hardening of the extensional viscosity was found to occur at earlier and earlier times, but always at roughly the same Hencky strain of $\varepsilon \approx 1.5$. In each case, at small Hencky strains, $\varepsilon < 1.5$, the Trouton ratio was found to start at values less than $\mathrm{Tr} < 10$ before strain hardening quickly in some cases to values as large as $Tr > 100$. Note that only in some cases of the virgin LDPE was the steady state value of the extensional viscosity obtained. For the case of both the toluene-treated LDPE and HDPE, the fluid filament was found to tear somewhere near the axial midplane before a steady state value could be obtained. As a result, the extensional viscosity data in Figs. 10 and 11 were found to reach a maximum extensional viscosity, but not necessarily achieve a steady-state value. To avoid confusion, we will designate the final value of the extensional viscosity or Trouton ratio in all cases as the maximum value even if that final value is in also the steady state.

A number of important differences in the extensional rheology of the virgin and toluene-treated samples can be observed from the data in Figs. 10 and 11. For the HDPE samples in Fig. 11, the differences are quite stark. Even at the highest extension rates of 2.0 s^{-1} little to no strain hardening was observed for the virgin HDPE even though the Weissenberg number is $\mathrm{Wi} = 0.6$. The toluene-treated HDPE is very different. For the toluene-treated HDPE, strong strain hardening with a maximum Trouton ratio of $Tr \approx 100$ was observed for extension rates as low as 0.6 s^{-1}. This rate corresponds to a Weissenberg number of $\mathrm{Wi} = 1.38$ which is more than $\mathrm{Wi} = 0.5$, where the transition to viscoelasticity and strain hardening is expected. This is likely because an averaged relaxation time (the disentanglement time) is being used to define the Weissenberg number here and the coil-stretch transition at $Wi = 0.5$ is defined based on the longest relaxation time of the fluid. Although the maximum Trouton ratio of the toluene-treated HDPE decreases with increasing extension rate and Weissenberg number as shown in Fig. 12, it is consistently one or two orders of magnitude larger than the virgin HDPE over the entire range of extension rates studied here. The difference between virgin and toluene-treated LDPE is not nearly as dramatic although the trends in Fig. 10 are consistent. As with the HDPE, an earlier onset of strain hardening for the toluene treated LDPE compared to the virgin LDPE was observed. This shift is consistent with the change in relaxation time observed in the shear rheology. Additionally, a modest increase in the maximum extensional viscosity and Trouton ratio was observed over most of the

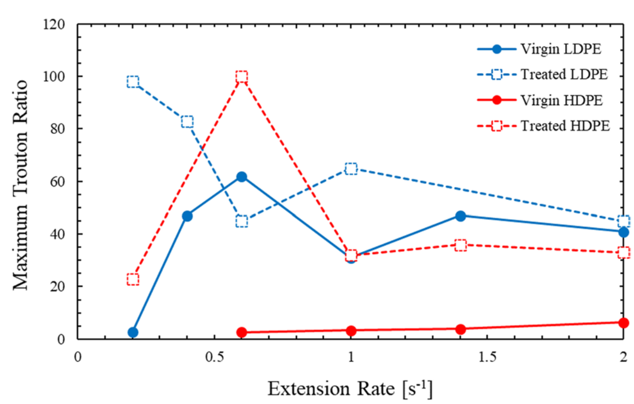

Fig. 12 Maximum Trouton ratios as a function of extension rate from uniaxial extensional rheology of the LDPEs and HDPEs. In blue is the LDPE and in red is the HPDE, filled circles correspond to virgin samples, and open squares are treated samples

data, although the increase was primarily observed at the lowest extension rates tested. This shift was also consistent with the difference observed in the shear rheology.

The dramatic increase in strain hardening of the extensional viscosity resulting from toluene treatment of the HDPE could have major implications on the utility of the polymer and the applications it can be used for following recycling. Perhaps even allowing the HDPE to be sold for applications like blow molding where strain hardening of the extensional viscosity is critical.

Conclusion

Key differences in the polyethylene samples before and after toluene treatment were noted. These differences will have a distinct effect not only on the performance of the final products but also during processing such as blow molding or injection molding into those products. A detailed analysis of the sample changes is critical to determine the viability of the recycling process. Wider usage of this technique beyond the early typically pilot scale commercialization, as mentioned in the introduction, has the potential to greatly reduce the amount of unrecycled multi-layer films currently impacting our environment. The primary finding of this work was that stabilizers, additives, and small molecular weight components were removed by solvent treatment. The impact of toluene treatment on thermal and rheological analysis was evident. During the first DSC heating ramp, the overall crystallinity of the toluene-treated PEs was substantially higher than that of the virgin samples (see Table 1). Time-resolved mechanical spectroscopy indicated an increase in viscosity and lower sample stability as a result of toluene treatment (see Fig. 7). The relaxation times for the toluene treated samples were higher causing a significant increase in the amount of strain hardening during the extensional rheology (see Figs. 9, 10, and 11). It is possible that the increase in the toluene-treated sample moduli, viscosity, and relaxation time was caused by crosslinking. However, little evidence of crosslinking was observed in the gas chromatography–mass spectrometry, gel permeation chromatography, or linear viscoelasticity measurements. An increase in the molecular weight distribution was not observed in the GPC data. Nor was a plateau or significant increase observed in the low frequency SAOS data of the LDPE at 175 °C or HDPE at 150 °C where the samples were shown to be thermally stable for extended periods of time. The increase in the moduli, viscosity, and relaxation time is therefore most likely the result of the loss of low molecular weight LDPE and HDPE fractions seen in the mass spec data. These fractions help plasticize the polymer melt, reducing its viscosity and making it more flowable. After toluene treatment, the samples appear more appropriate for blow molding or other applications where larger shear and extensional viscosities are needed when compared to the virgin samples. In a more general sense, the treated samples may not have the same application as the virgin counterpart based on the material property changes or simply may require the reintroduction of low molecular weight fractions and stabilizers to reach the same markets.

While the nominal melt temperature, defined by the DSC peak, is typically taken from the second heating ramp, it is noteworthy that the first heat crystalinity is what will be present during processing. Solvent treatment at 110 °C, below the nominal PE melt temperature, will cause the dissolution of the low melting fraction of the crystallized LDPE. The "surviving" crystals after toluene treatment could act as nucleation sites when crystallizing again. Furthermore, residual solvent still present during the first DSC heating might increase chain mobility but is expected not be present for the second DSC cycle, since solvent will evaporate during the first DSC cycle. The risk of residual solvent being present could be an industry concern and may necessitate a pre-treatment step to guarantee its removal.

The difference in stability between the virgin and toluene-treated LDPE and HDPE is likely the result of the toluene dissolving and stripping out any stabilizers that were added to the virgin LDPE and HDPE samples during their original manufacturing. The ability of solvent dissolution to strip additives has been of distinct interest within the industry, predominantly as a means to remove dyes or harmful additives (Ügdüler et al. 2020). The recycled polymers would potentially need to have stabilizers reintroduced to meet consumer requirements. Without this pre-processing step, our experiments clearly demonstrate that STRAP could have a detrimental effect on the stability of HDPE and LDPE. It is necessary to study the stability of the STRAP recycled polymers before marketing them.

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Data availability The authors confirm that the data supporting the findings of this study are available within the article [and/or] its supplementary materials.

Declarations

Competing interests This material is based upon work supported by the US Department of Energy, Office of Energy Efficiency and Renewable Energy, Bioenergy Technologies Office under award number DEEE0009285.

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